

# Malhar Paresh Mangtani

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## Education

- Columbia University** New York, NY  
**M.S. in Computer Science** Aug 2026 – Dec 2027
- Relevant Coursework: Analysis of Algorithms, Artificial Intelligence, Deep Learning for Computer Vision, Introduction to Databases.
- Sarvajanik College of Engineering and Technology** Surat, IN  
**B.Tech in Computer Engineering**, CPI: **9.35/10.00**, Degree CGPA: **9.11/10.00** Sep 2022 – May 2026
- Relevant Coursework: Machine Vision, Python for Data Science, Data Structures & Algorithms, Database Management Systems, Operating Systems, Cloud Computing, Compiler Design.
- Sarvajanik College of Engineering and Technology** Surat, IN  
**Honors in Artificial Intelligence & Machine Learning**, CGPA: **10.00/10.00** Dec 2023 – Dec 2025
- Relevant Coursework: Introduction to Artificial Intelligence and Robotics, Machine Learning and Applications, Deep Learning, Modern trends in AI-ML.

## Work Experience

- Oklahoma State University, Industrial Training and Assessment Center (ITAC)** Stillwater, OK  
**AI Intern** Feb 2026 – Apr 2026
- Engineered SmartITAC — an AI chatbot enabling the ITAC team to query 167,000+ energy assessment records through plain English, addressing a significant operational need while reducing a 1–2 hour task to seconds.
  - Implemented an end-to-end RAG pipeline using HuggingFace embeddings, ChromaDB, and Groq API with a multi-mode query system supporting keyword search, geographic filtering, statistical aggregation, and time-based filtering.
  - Developed a FastAPI backend with real-time streaming and an HTML chat frontend accessible to all team roles.
  - Recognized by Dr. Hitesh D. Vora, Director of Oklahoma State University's ITAC, for exceptional contributions to the design and development of SmartITAC.
- USICAMM, Secretaría de Educación Pública (SEP)** Mexico, MX  
**AI Intern** Jan 2025 – Jul 2025
- Designed AI-based background verification and profile evaluation modules for teacher registration and promotion.
  - Built an AI-based validation search engine supporting teacher onboarding and management across Mexico.
- Laurentian University** Sudbury, CA  
**AI/ML Research Intern** Jul 2024 – Aug 2024
- Trained a DeepLabV3 + ResNet50 model for automated nuclei segmentation in histopathology images using Python and TensorFlow.
  - Achieved **95.55%** validation accuracy, **98.75%** Dice, and **97.58%** Jaccard Index over **10** epochs, demonstrating strong segmentation performance.

## Projects

- Synthetic CT Generation from MRI** 2025
- Designed and iteratively improved a 3D deep learning pipeline for MRI-to-CT synthesis, progressing from a 3D Pix2Pix GAN (3D U-Net generator) to an anatomy-aware SwinGAN (SwinUNETR generator), both paired with a ConvNeXt-based anatomical classifier.
  - Achieved stronger results with SwinGAN across all anatomical regions (avg. MS-SSIM **0.932**, PSNR **27.83** dB, MAE **0.027**), with peak performance on Thorax (MS-SSIM **0.947**, PSNR **28.30** dB, MAE **0.026**), outperforming the Pix2Pix baseline with high structural fidelity in complex bone-air interface regions. Published in the Proceedings of IEEE ICETCI 2026, Hyderabad, India.
- Malaria Parasite Detection** 2025
- Implemented an automated malaria parasite detection system using pre-trained ConvNeXt with Grad-CAM++ for explainable predictions on 27,560 NLM-Malaria images.
  - Achieved **99%** training and **98%** test accuracy, outperforming five baseline CNN models.
  - Published and presented at IEEE ICETCI 2025, Hyderabad, India; received the Best Presenter Award.
- Emotion Detection Using Deep Learning Algorithms** 2025
- Evaluated multiple CNN architectures (ResNet50, EfficientNetB3, VGG19) for a 6-class image classification task.
  - Achieved the highest performance with ResNet50, reaching **93.36%** validation accuracy and **67.22%** test accuracy, supported by confusion matrices and detailed classification reports.

## Technical Skills

- Programming:** Python, Java, SQL
- Software & Tools:** Git/GitHub, FastAPI, VS Code, Jupyter Notebook, Conda
- AI/ML:** PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, RAG, LLMs, OpenCV, NumPy, Pandas
- ML Models & Methods:** CNNs, RNNs, LSTMs, GANs, Transformers, Transfer Learning, Model Evaluation